Project Design Phase–II

Data Flow Diagram & User Stories

Date	25.09.2026	**Team ID**	SWTID-2026-4307
Project Name	AI Weather-Wise API	**Maximum Marks**	4 Marks

Data Flow Diagrams

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: DFD Level 0 (Industry Standard) & System Architecture Flow

1. User configures microclimate location / field coordinates and request parameters in client app / dashboard.
2. User sends request to AI Weather-Wise API gateway.
3. API fetches live satellite & sensor data from the core microclimate engine.
4. AI Model processes data and generates ML predictions & automated risk alerts.
5. Analyzed response is presented to the user via JSON API / Dashboard.

User Stories

Use the below template to list all the user stories for the product:

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user / Farmer)	Registration & Authentication	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	I can access my account / dashboard upon successful registration.	High	Sprint-1
Customer (Mobile user / Farmer)	Registration	USN-2	As a user, I will receive a confirmation email once I have registered for the application.	I can receive confirmation email & click confirm to verify account.	High	Sprint-1
Customer (Mobile user / Event Organizer)	OAuth Registration	USN-3	As a user, I can register for the application through Facebook / Google login.	I can register & access the dashboard with Facebook / Google Login.	Low	Sprint-2
Customer (Mobile user / Event Organizer)	Gmail Registration	USN-4	As a user, I can register for the application through Gmail.	I can register & access the dashboard with Gmail credentials.	Medium	Sprint-1
Customer (Mobile user	Login & Authentication	USN-5			High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
/ Event Organizer)			As a user, I can log into the application by entering my registered email & password.	I can successfully log in and view my personalized weather dashboard.		
Customer (Web user / Developer)	API Key Management	USN-6	As an API consumer, I can generate, view, and regenerate my API access keys from the developer dashboard.	I can securely copy API keys and authenticate requests to the AI Weather-Wise endpoints.	High	Sprint-1
Customer (Web / Mobile user)	AI Weather Forecasting & Microclimate Alerts	USN-7	As a user, I can query hyper-local weather predictions and receive AI-driven automated risk alerts.	I can view real-time microclimate predictions and receive push/email notifications for sudden weather shifts.	High	Sprint-2
Customer Care Executive	Support & Ticket Resolution	USN-8	As a customer care executive, I can view user support tickets regarding API downtime or weather data inaccuracies.	I can respond to and resolve user tickets efficiently within the support portal.	Medium	Sprint-2
Administrator	System Analytics & User Management	USN-9	As an administrator, I can monitor overall API usage metrics, user management, and AI engine model performance.	I can view system logs, manage active user accounts, and track endpoint performance metrics.	High	Sprint-2